Research Protocol

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01 Title

A Virtue-Ethics-Based Reflective Scaffolding Agent for the Responsible Use of Generative AI by University Students: Annotated Dataset, Conversational Artifact, and Mixed-Methods Evaluation

02 Abstract

Generative AI makes full delegation of academic work easy, and institutional responses have focused on restricting it. This Design Science Research project, grounded in pragmatism, will design and evaluate a conversational agent that scaffolds virtue-based deliberation (prudence and academic honesty) when Peruvian undergraduates ask generative AI to do their work. It will build a Spanish-language corpus of academic-integrity scenarios annotated for the four cardinal virtues plus honesty (Krippendorff's alpha per virtue), derive the agent's scaffolding policy from it, and compare the agent with unrestricted and restrictive-normative conditions in a sequential explanatory mixed-methods quasi-experiment with intact course sections. Outcomes are metacognitive reflection, AI dependence and product integrity, including performance without the tool. A rapid review of 10 comparative studies found no agent that scaffolds the delegation decision itself or uses ethics as a design basis. Expected contributions are evaluated design principles (a nascent design theory) and an open Spanish-language resource, subject to ethics approval.

03 Introduction & Problem Statement

Generative AI (GenAI) has entered Peruvian undergraduate coursework faster than institutions have developed pedagogical responses to it. A survey across four Peruvian universities associated ChatGPT use with students' ethical concerns and called for responsible-use programmes (Acosta-Enriquez et al., 2024). In my own courses I observe a recurring pattern of full delegation: students paste the assignment into a chatbot and submit the output without mediating it through their own judgment.

Institutional responses have concentrated on restricting conduct — detectors, prohibitions, sanctions. Restriction may reduce visible misuse, but it does not build the capacity to decide when delegation is legitimate, which is precisely what students will need when restrictions are absent. The underexplored alternative is to form judgment at the moment of delegation. Virtue ethics offers a theory of such judgment (practical wisdom), but, in the studies reviewed in Deliverable 04, it has not been operationalised as a design basis for educational AI, nor evaluated against restrictive approaches, nor supported by Spanish-language resources.

Problem statement. It is not known how to design a conversational agent that scaffolds virtue-based deliberation when undergraduates delegate work to GenAI, nor whether forming judgment in this way outperforms restricting behaviour on metacognitive reflection, product integrity and AI dependence.

04 Literature Review

A rapid review reported with PRISMA 2020 (Deliverable 04; Page et al., 2021) searched Semantic Scholar on 24 October 2026 (617 records) and included 10 comparative studies of generative-LLM tools designed to elicit higher-education students' own reasoning. Three findings frame this protocol.

First, tool-level scaffolding (hints instead of answers, Socratic questioning, reflection or metacognitive prompts) improved process outcomes such as self-regulation and argument quality (Lee et al., 2024; Yang et al., 2025; Aldamen et al., 2026), and personalised LLM coaching lowered self-reported AI dependency relative to template prompts (Yuan, 2026).

Second, effects on objective performance were mostly positive (7 of 10 studies), but two found no gain in unassisted performance (Kumar et al., 2024; Yuan, 2026), consistent with a large field trial in secondary mathematics in which a Socratic tutor prevented the harm of unrestricted access but produced no net learning (Bastani et al., 2025). Only one included study tested delayed transfer without AI (Aldamen et al., 2026); this protocol therefore adds an unassisted post-test.

Third, the evidence comes mainly from Asia (6 of 10 studies) and from language and computing courses (7 of 10); several studies do not report the model or its configuration; only 5 of 10 compared the tool against a plain LLM, so this protocol includes an unrestricted condition; and no included study scaffolds the decision of whether to delegate or uses ethics as a design basis. These knowledge, methodological, contextual and theoretical gaps define RQ1 and RQ2. The review was capped at 10 studies; absence claims are bounded by it.

05 Questions, Hypotheses & Objectives

RQ1 (design). How can a conversational agent grounded in virtue ethics scaffold ethical deliberation when undergraduates at a Peruvian university delegate work to GenAI?

RQ2 (evaluation). Does forming autonomous judgment through the agent outperform restricting behaviour on metacognitive reflection, product integrity and AI dependence, relative to unrestricted and restrictive-normative conditions?

Hypotheses *(RQ2 only; RQ1 yields design propositions, not tests).*

- H1: the agent condition produces higher coded levels of reflection than both comparison conditions.
- H2: the agent condition produces lower AI dependence than the unrestricted condition; the contrast with the restrictive condition is non-directional, because restriction may lower dependence in the short term.
- H3: the agent condition produces higher product integrity than the unrestricted condition and at least equivalent integrity to the restrictive condition.

General objective. To design, build and evaluate a virtue-based reflective scaffolding agent and derive design principles for the responsible use of GenAI in higher education.

Specific objectives.

- SO1: build and validate a Spanish-language corpus of academic-integrity scenarios involving GenAI, annotated for the four cardinal virtues plus honesty.
- SO2: design and implement the agent and two comparison conditions (unrestricted; restrictive-normative).

- SO3: evaluate the agent on the three outcomes through a sequential explanatory mixed-methods design.
- SO4: formulate evaluated design principles and adoption guidelines for higher-education institutions.

06 Methodology

Philosophical stance and approach. Pragmatism (Goldkuhl, 2012): knowledge is warranted by its usefulness in guiding action. Design Science Research is the research approach (Hevner et al., 2004), and the intended contribution is a nascent design theory (Gregor & Hevner, 2013).

Artifacts. (a) Annotated corpus: scenarios validated by experts; annotation anchored in exemplar cases rather than rule-based definitions, since practical wisdom resists codification; agreement reported as Krippendorff's alpha per virtue, not aggregated (Hayes & Krippendorff, 2007). (b) Agent: an LLM-based conversational artifact whose prompting implements Socratic, virtue-oriented scaffolding, informed by the corpus.

Evaluation strategy. FEDS, Human Risk & Effectiveness (Venable et al., 2016). Formative/artificial episodes: expert validation of virtue definitions and pilot annotation. Summative/naturalistic episode: three-condition comparison in real course sections.

Design. Sequential explanatory mixed methods (Ivankova et al., 2006).

- **Quantitative strand:** quasi-experiment with pre/post measures; the intact section is the unit of assignment, with at least two sections per condition so that condition is not confounded with a single section; multilevel models with section as cluster; sample size set by a priori power analysis including the design effect; a manipulation check verifies that the restrictive condition is perceived as restrictive.
- **Qualitative strand:** case-anchored analysis of conversation logs, with divergent and extreme cases selected from the quantitative results; integration through joint displays (Fetters et al., 2013).

Operationalisation of outcomes

Outcome	Instrument / measure	Status in Spanish
Metacognitive reflection	Coding of student reflections and logs with the four-category scheme: non-reflection, understanding, reflection, critical reflection (Kember et al., 2008)	Requires translation, expert review and pilot agreement check
AI dependence	Scale for Dependence on AI (DAI), unifactorial (Morales-García et al., 2024)	Developed with Peruvian university students (N = 528)
Product integrity	Composite: (a) provenance ratio — share of submitted text traceable verbatim to AI output in the logs; (b) blinded expert rubric on disclosure and own contribution	No established instrument; content validity by expert judgment to be reported

07 Ethical Considerations

I will be designer, instructor and evaluator of my own sections. Mitigations:

- Approval by the institutional research ethics committee of the host university before any data collection.
- Consent obtained by a third party and decoupled from grading: refusal or withdrawal has no academic consequence, and I will not know who consented until grades are closed.
- Outcome coding by blinded external coders.
- Logs pseudonymised at capture, kept on institution-controlled storage (never public repositories), with defined retention and deletion.
- No student data sent to third-party model providers without a data-processing agreement (the agent will use a commercial model through an API, so this agreement is required).
- Disclosure of AI assistance in all research outputs, consistent with the object of the thesis.

08 Expected Results

- Evaluated virtue-based design principles for reflective scaffolding agents (the nascent design theory), instantiated in a documented, replicable agent.
- A Spanish-language annotated corpus with annotation guide and per-virtue agreement, released openly if ethics approval permits.
- Comparative evidence on the three outcomes, reported whether or not it supports the hypotheses.
- Venues: Computers and Education: Artificial Intelligence; International Journal for Educational Integrity; DESRIST; IEEE-RITA.

09 Timeline & Budget

Period	Phase (DSR activity)	Main output
2026-II – 2027-I	Problem, literature, ethics approval	Approved protocol; literature review
2027-I – 2027-II	Corpus construction and annotation (formative)	Validated corpus; Paper 1 (dataset)
2027-II – 2028-I	Agent design and formative evaluation	Agent v1; pilot and manipulation check
2028-I – 2028-II	Summative quasi-experiment + qualitative strand	Outcome data; coded logs
2028-II – 2029-I	Integration, design principles, writing	Paper 2 (evaluation); thesis

Budget (to be costed): LLM API inference (commercial provider under a data-processing agreement); external annotators, expert validators and blinded coders; ethics fees if applicable; open-access charges. Funding: self-funded; CONCYTEC/ProCiencia calls to be assessed.

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